

Solana Fit Check

Every rule and feature from Parts 3 and 4 of *What Won, and How We Win*, checked against the Solana Field Guide 2026: what Solana already gives us, what we assemble from its parts, what it lacks, and how we fill each gap.

14

On Solana

Solana has it today. We use it directly.

11

Parts exist

Solana has the building blocks. We wire them together, with some of our own code.

2

Not on Solana

No Solana feature does this. We fill the gap ourselves.

5

Our design

Not a chain question. It is how we build the product.

32 items checked. 25 of them run on Solana features today, 2 need something Solana does not provide, and 5 are product decisions, not chain features.

The short version

- **The core of our plan is native to Solana:** the cap enforced by the token program (B2, X1), one-tap Blink approvals (B7), memo receipts (B6), owner revoke (B8) and x402 payments (B5).
- **Two real gaps:** a budget tree with ancestor debit (B1) and a receipt per request (X4). B1 is solved in core/ now; X4 is a small program on the roadmap.
- **Quickest wins we are not using yet:** Kora (agents pay fees in USDC, never hold SOL) and the Solana Attestation Service (seller reputation for X9).
- **Check first on devnet:** whether @x402/svm accepts a delegate as payer (B5) and whether Squads spending limits work on devnet (X10).

What Solana does not have, and how we solve it

● Not on Solana

Budget tree with ancestor debit

Helpers get less than their parent; every spend counts all the way up.

How we solve it: No Solana feature does a tree. Hackathon: core/ keeps the tree, and each agent's on-chain delegate amount is reset to min(its own remaining, every ancestor's remaining) after each spend, so the chain cap is never looser than the tree. Later: a small program with a PDA per node.

● On Solana

Real payments, verified from the chain

x402 in USDC on devnet; settled only after reading it back.

How we solve it: Use 'exact' pricing. Kora lets agents hold only USDC, no SOL, matching what Cordon praised about Arc. Verify Friday: does @x402/svm accept a delegate as the payer?

Verify: x402 + delegate payer untested; 'upto' still listed as EVM-only.

● Not on Solana

Receipt per request (no double charge)

A receipt account keyed by request hash; paying twice is impossible.

How we solve it: No ready program. Write a small Anchor program: create PDA(request hash) + transfer in one transaction; a second attempt fails because the PDA exists. Roadmap; compression makes it cheap at scale.

● Parts exist

Drain-proof wallet (discussed, not yet in the doc)

Thief can't pay themselves; a small bucket refills slowly; refills stop on attack.

How we solve it: Recipient list and period cap via Squads. A capped refilling bucket does not exist: Streamflow streams keep accruing if unused, so the cap needs our refill service or a small program. We cannot freeze USDC accounts (Circle holds the freeze authority), so 'freeze' means revoking the delegate, via a Blink to the owner or a Squads guardian member.

Verify: Squads spending limits on devnet untested.

Item by item

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
R1	Enforce where the money moves Limits checked by the program that moves funds, not only by our app.	LIVE SPL Token approve / revoke (P-Token) STANDARD Squads spending limits STANDARD Token-2022 TransferHook DEVNET Fixed-delegation program	● On Solana	The token program itself refuses a delegate transfer above the approved amount. Squads adds periods and allowed recipients.
R2	The model never holds the key A separate signer; the AI can only ask.	LIVE Keychain (local keys, KMS, HSM, wallets) STANDARD Embedded wallets with key policies (Privy, Turnkey) LIVE Solana Agent Kit v2 (scoped keys, caps)	● Parts exist	Custody tools exist. Keeping the key out of the model's process is our design: a signer process with Keychain, and an AI process with an empty environment (Capsule pattern).
R3	The agent can't choose who gets paid or how much Price and payee come from the 402 or the chain and must match policy.	LIVE x402 facilitator (@x402/svm) STANDARD Squads allowed destinations	● Parts exist	x402 puts payTo and amount in the 402. Our gate reads them (never from the agent) and matches them against policy; Squads destinations make the chain refuse other recipients too.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
R4	Refuse before signing, with reason codes Checks in order; a closed list of reasons the agent can act on.	STANDARD simulateTransaction (RPC) STANDARD Program custom errors	Parts exist	The gate is ours (core/ codes). Solana helps: simulate a payment for free before signing to catch an on-chain refusal early.
R5	Fail closed Unknown, unreadable or missing means no.	STANDARD Atomic transactions (all or nothing)	Our design	A product rule we apply everywhere. Solana's all-or-nothing transactions mean a failed check never leaves a half-done payment.
R6	Don't trust the facilitator; read the chain A payment counts as settled only after we read it back.	STANDARD RPC getTransaction STANDARD RPC WebSocket subscriptions STANDARD Confirmation levels COMING Alpenglow ~150 ms finality	On Solana	Read every payment back with getTransaction at 'confirmed'. Alpenglow (expected Oct 2026) would make this near-instant.
R7	Human approval bound to the exact purchase Approve exactly this payment, not a general permission.	STANDARD Actions & Blinks STANDARD Passkeys (Secp256r1) STANDARD Squads approvals STANDARD Partial signing	On Solana	A Blink hands the owner the exact transaction to sign, so the approval is bound to it by construction. Passkeys let a phone sign it.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
R8	Kill switch that doesn't need our server Owner stops an agent directly on chain.	STANDARD SPL revoke STANDARD Squads: remove spending limit	● On Solana	The owner signs one revoke from their own wallet. No server involved.
R9	Limit the blast radius in time Budgets per period; future budget can't be stolen now.	STANDARD Squads spending limits with periods DEVNET Fixed-delegation program (amount + expiry) LIVE Streamflow per-second streams STANDARD Subscriptions via delegation	● Parts exist	Period budgets exist (Squads). NotYet's time-locked keys have no Solana equivalent; if we want them, drand time-lock encryption (tlock-js) works off chain and is chain-agnostic.
R10	Public audit trail Every payment, refusal, grant and revoke publicly readable.	STANDARD Payments with memos STANDARD RPC + Geyser streaming LIVE Solana Attestation Service	● On Solana	Memos on payments, plus a memo-only transaction for refusals and revokes. No dedicated log service like Hedera HCS is needed.
R11	Prove it with an experiment Attack runs, with-vs-without tests, reproducible results.	LIVE Surfpool (local validator) STANDARD LiteSVM & Mollusk test harnesses STANDARD Verified builds	● On Solana	Run the red-team attack thousands of times in LiteSVM, then live on devnet. Verified builds prove deployed code matches the repo.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
R12	Label what is mocked Say exactly what is real.	No chain feature needed	● Our design	README and demo labels. Use devnet explorer links for everything that is real.
R13	Free to look, pay only when needed Discovery free; never pay for what can't be delivered.	STANDARD RPC reads and simulation (free) STANDARD Delivery vs Payment	● Parts exist	Delivery vs Payment is atomic only for on-chain assets. For API data, settle after a valid response (TrueTick pattern) or use an escrow PDA (X4).
R14	Legible to agents and judges MCP, OpenAPI, llms.txt, one-command setup.	LIVE Solana Agent Kit v2 LIVE x402 packages STANDARD IDL & Codama client generation STANDARD AI tools docs	● Parts exist	Agent tooling exists; our MCP server, llms.txt and one-command demo are ours.
B1	Budget tree with ancestor debit Helpers get less than their parent; every spend counts all the way up.	STANDARD SPL approve (one delegate per account) STANDARD Squads spending limits (flat, per member) DEVNET Fixed-delegation (single level)	● Not on Solana	No Solana feature does a tree. Hackathon: core/ keeps the tree, and each agent's on-chain delegate amount is reset to min(its own remaining, every ancestor's remaining) after each spend, so the chain cap is never looser than the tree. Later: a small program with a PDA per node.
B2	Cap enforced by Solana itself Over the cap, the chain refuses.	LIVE SPL approve / revoke (P-Token, 95-98% fewer CU) STANDARD Squads spending limits	● On Solana	One token account per agent with an approved delegate. Note: only one delegate per token account at a time.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
B3	Refuse-before-sign gate Ordered checks with reason codes before any signature.	STANDARD simulateTransaction	● Our design	Our gate in front of the signer, with core/ reason codes. Simulation catches chain-level refusals before sending.
B4	The AI never holds a key; no pay tool Agent's only tool is fetch_paid(url, purpose).	LIVE Keychain STANDARD Embedded wallet policies	● Parts exist	Signer behind Keychain (local key now, KMS later). The tool design is ours.
B5	Real payments, verified from the chain x402 in USDC on devnet; settled only after reading it back.	LIVE x402 facilitator (@x402/svm, PayAI, Coinbase CDP; 'exact') LIVE Kora fee abstraction (pay fees in USDC) STANDARD getTransaction	● On Solana	Use 'exact' pricing. Kora lets agents hold only USDC, no SOL, matching what Cordon praised about Arc. Verify Friday: does @x402/svm accept a delegate as the payer? Verify: x402 + delegate payer untested; 'upto' still listed as EVM-only.
B6	Receipt on every payment Each payment carries budget id and purpose hash.	STANDARD Payments with memos STANDARD Token-2022 MemoTransfer (requires memos)	● On Solana	Add a memo instruction to every payment. MemoTransfer only applies to Token-2022 accounts; USDC is classic SPL, so we add the memo ourselves.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
B7	One-tap approval for big purchases Owner approves exactly one purchase from a phone.	STANDARD Actions & Blinks STANDARD Passkeys	● On Solana	A Blink per held purchase, unfurling in chat or on X.
B8	Kill switch without our server Owner revokes in one transaction; other agents keep working.	STANDARD SPL revoke	● On Solana	Owner signs revoke on that agent's token account only.
X1	“Turn off our server” With our backend off, a compromised agent still can't overspend.	STANDARD SPL delegate cap in the token program	● On Solana	Works out of the box: the delegate cap lives in the token account. This is our clearest 'why Solana'.
X2	Red-team arena Hostile agent tries to drain; scoreboard of attempts vs losses.	STANDARD LiteSVM / Surfpool STANDARD Devnet	● Parts exist	Thousands of attack runs locally in LiteSVM (Cordon-style numbers), one live run on devnet.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
X3	Blast-radius meter The most an attacker could take right now.	STANDARD getTokenAccountsByOwner (delegated amount per account) STANDARD RPC WebSocket	● On Solana	Sum the delegated amounts across agent accounts; update live on each change.
X4	Receipt per request (no double charge) A receipt account keyed by request hash; paying twice is impossible.	STANDARD PDAs PARTIAL Reduced rent (~90%) LIVE ZK Compression (rent-free accounts)	● Not on Solana	No ready program. Write a small Anchor program: create PDA(request hash) + transfer in one transaction; a second attempt fails because the PDA exists. Roadmap; compression makes it cheap at scale.
X5	Money that carries its own rules Every transfer checks sellers and limits, wherever it goes.	STANDARD Token-2022 TransferHook STANDARD Token ACL (allow / block lists)	● Parts exist	USDC is classic SPL, so hooks don't apply to it. We would wrap USDC into a Token-2022 mint with a hook. Roadmap.
X6	Chain-built invoice Report from memos, each line linked to its transaction.	STANDARD Memos STANDARD RPC / Helius indexing	● On Solana	Read the agent accounts' history, parse memos, group by budget.

#	ITEM	SOLANA FEATURES (STATUS)	VERDICT	HOW WE USE IT / HOW WE SOLVE IT
X7	Don't pay twice + savings number Reuse what the team already bought.	No chain feature needed	● Our design	A lookup in our server keyed by request hash; the hash also goes in the memo.
X8	Spending advisor Suggest revokes and smaller caps from spending history.	STANDARD RPC data	● Our design	LLM over computed figures (LedgerMind pattern: do the math in code first).
X9	Trust ramp for new sellers Trial limit for new sellers; track record anyone can rebuild.	STANDARD Memos LIVE Solana Attestation Service STANDARD Squads destinations	● Parts exist	Outcomes in memos; publish seller scores as Solana Attestation Service credentials so any app can reuse them. Solana has no agent reputation registry (EVM has ERC-8004); SAS is the closest fit.
X10	Drain-proof wallet (discussed, not yet in the doc) Thief can't pay themselves; a small bucket refills slowly; refills stop on attack.	STANDARD Squads destinations + periods LIVE Streamflow per-second streams STANDARD RPC WebSocket / Geyser STANDARD SPL revoke	● Parts exist	Recipient list and period cap via Squads. A capped refilling bucket does not exist: Streamflow streams keep accruing if unused, so the cap needs our refill service or a small program. We cannot freeze USDC accounts (Circle holds the freeze authority), so 'freeze' means revoking the delegate, via a Blink to the owner or a Squads guardian member. Verify: Squads spending limits on devnet untested.

Solana features our plan doesn't use yet

From the field guide. Each could strengthen the plan; Kora and the Attestation Service are the quickest wins.

FEATURE	STATUS	WHAT IT WOULD GIVE US
Kora fee abstraction	LIVE	Agents pay fees in USDC and never hold SOL. Removes a setup step and matches Arc's 'one asset' story.
Solana Attestation Service	LIVE	Reusable credentials: seller reputation (X9), verified-owner badges, KYC for business customers.
Confidential balances	STANDARD	Hide spend amounts from competitors with an auditor key for the accountant. Token-2022 only, so with wrapped USDC.
ZK Compression	LIVE	Rent-free receipt accounts at scale for X4.
Streamflow	LIVE	Per-second budget streams; a ready-made drip for period budgets.
Passkeys (Secp256r1)	STANDARD	Owner approves with Face ID / fingerprint, no wallet app.
Fixed-delegation program	DEVNET	Capped, expiring agent budgets already on devnet. Could replace our own delegate handling, but maintainer unclear and devnet only.
P-Token	LIVE	Token transfers 95-98% cheaper in compute; makes per-call payments even cheaper.
Alpenglow	COMING	~150 ms finality expected Oct 2026; settlement checks become near-instant.
Larger transactions (4 KB)	LIVE	Payment + memo + receipt instructions fit in one transaction.

Tools the ETHOnline winners used that Solana lacks

The winners built on Arc, Hedera, Sepolia and Sei. Where their tools have no Solana twin, this is what we use instead.

THEIR TOOL	ON SOLANA	OUR SUBSTITUTE
ENSv2 per-record permissions (Capsule, Harness, Batas)	SNS .sol names resolve to wallets but have no per-record write roles.	Keep permissions in Squads or our own program; use .sol only as a readable name.
ERC-8004 agent identity and reputation	No standard agent registry on Solana.	Solana Attestation Service credentials for agent and seller identity.
Hedera Consensus Service topics (public logs)	No dedicated log service.	Memo transactions or program events; index with RPC or Geyser.
EIP-3009 gasless authorisations	No signed-authorisation transfer standard.	Kora pays fees; partial signing lets a fee payer co-sign.
USDC as gas (Arc)	Fees are in SOL.	Kora fee abstraction lets users pay in USDC.
Uniswap v4 hooks / SwapVM opcodes (Batas)	Different model.	Token-2022 TransferHook for transfer-level rules.
drand time-lock keys (NotYet)	Not a chain feature anywhere.	tlock-js works off chain with any chain.

LEGEND AND SOURCES

- On Solana Solana has it today. We use it directly.
- Parts exist Solana has the building blocks. We wire them together, with some of our own code.
- Not on Solana No Solana feature does this. We fill the gap ourselves.
- Our design Not a chain question. It is how we build the product.

Feature statuses come from the Solana Field Guide 2026 (compiled 2 Oct 2026): LIVE on mainnet, STANDARD long-standing, DEVNET test networks only, PARTIAL partly activated, COMING scheduled. Items marked "verify" are untested assumptions to check on devnet first.

Plan items come from Analysis/what-won-and-how-we-win.pdf Parts 3 and 4. X10 is the drain-proof wallet discussed after that document and is not in it yet.